

ASCII MASTER FLOW – PANEL 1/12: Cell and organelle map

PROKARYOTE -> no membrane-bound nucleus or organelles

EUKARYOTE -> nucleus + compartmental organelles

RIBOSOME -> protein | MITOCHONDRION -> ATP

ER + GOLGI -> processing and movement

PLANT -> cellulose wall + chloroplast + large vacuole

ASCII MASTER FLOW – PANEL 2/12: Membrane transport decision tree

SOLUTE DOWN GRADIENT -> diffusion

WATER + SELECTIVE MEMBRANE -> osmosis

AGAINST GRADIENT + ENERGY -> active transport

EXTERNAL TONICITY -> cell water response

TRAP -> movement depends on membrane and gradient

ASCII MASTER FLOW – PANEL 3/12: Cell division comparison

MITOSIS -> one division -> two similar cells

ROLE -> growth and repair

MEIOSIS -> two divisions -> four haploid products

ROLE -> gametes + variation

CROSSING OVER != gene editing

ASCII MASTER FLOW – PANEL 4/12: Gene-expression rail

DNA --replication--> DNA

DNA --transcription--> mRNA

mRNA + tRNA + RIBOSOME --translation--> protein

microRNA -> post-transcriptional regulation

PROTEIN -> cell structure / enzyme / receptor / signal

ASCII MASTER FLOW – PANEL 5/12: Nutrition and digestion route

MOUTH -> amylase

STOMACH -> acid + pepsin

LIVER / BILE -> fat emulsification

PANCREAS + INTESTINE -> digestion completion

SMALL INTESTINE -> principal absorption

ASCII MASTER FLOW – PANEL 6/12: Transport respiration excretion loop

LUNGS -> alveolar gas exchange

BLOOD -> gases + nutrients + hormones + wastes

HEART -> pulmonary + systemic circulation

CELLS -> ATP-producing respiration

NEPHRON -> filtration + reabsorption + secretion

ASCII MASTER FLOW – PANEL 7/12: Control and feedback ladder

RECEPTOR -> change detected

NERVOUS -> rapid wired response

ENDOCRINE -> blood-borne hormone response

TARGET -> physiological adjustment

NEGATIVE FEEDBACK -> restores regulated range

ASCII MASTER FLOW – PANEL 8/12: Immunity architecture

BARRIERS + PHAGOCYTES + COMPLEMENT -> innate

B CELL -> antibody / memory

T CELL -> helper / cytotoxic / regulatory roles

REGULATORY T CELL + Foxp3 -> peripheral tolerance

SELF-REACTIVITY CONTROL != no immune response

ASCII MASTER FLOW – PANEL 9/12: Disease transmission grid

BACTERIUM | VIRUS | FUNGUS | PROTOZOAN | HELMINTH

AIR | WATER/FOOD | BLOOD | CONTACT | VECTOR

VECTOR -> carries pathogen, is not disease

ZOONOSIS -> animal-human interface

PREVENTION -> mechanism-specific layers

ASCII MASTER FLOW – PANEL 10/12: Microbial ecology and resistance

ANTIMICROBIAL PRESSURE -> selection

RESISTANT MICROBE -> survival + spread

BIOFILM -> matrix-protected community

MICROBIOME / PROBIOTIC -> context-dependent roles

EXTREMOPHILE LABEL -> tolerance, not universal safety

MUST REMEMBER: Biology and physiology connect cell structure, metabolism, genetics,...

ASCII MASTER FLOW – PANEL 11/12: Plant function flow

ROOT -> water + mineral uptake

XYLEM -> upward transport

LEAF -> stomata + photosynthesis + transpiration

PHLOEM -> food translocation

C3 | C4 | CAM -> different carbon-fixation strategies

CLOSE DISTINCTION: Gene is not allele or genome, genotype is not phenotype, pathogen is...

ASCII MASTER FLOW – PANEL 12/12: PYQ and evidence-status rail

TAXON / PLANT FORM / SPECIES PROPERTY -> exact classification

CELLULOSE -> cellulase -> sugars -> microbial metabolism

RESEARCH -> SURVEILLANCE -> PROGRAMME DELIVERY -> REGULATION

WHO VALIDATION != pathogen eradication

MECHANISM / ASSOCIATION / APPROVAL / OUTCOME -> separate rungs

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name organism/cell/tissue/organ,...